

Reproduce-then-Extend — State Welfare Spending (American State Politics) (Day 1)

Intro to Machine Learning · Bruce A. Desmarais

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Published study. The determinants of **state welfare effort** are a long-running question in American state politics and policy (e.g., Plotnick & Winters, “A Politico-Economic Theory of Income Redistribution,” *APSR* 1985; Brown, “State Responses to the Welfare Crisis,” 1995). The canonical design is a **cross-sectional OLS** of per-capita state welfare spending on economic need, fiscal capacity, political ideology, and organized labor. This example illustrates **regularization and variable selection** in a small-sample ($n = 50$) design.

1 Background

Why do some states spend far more on social welfare than others? The state welfare-effort literature points to **fiscal capacity** (richer states can afford more), **need** (poverty and inequality), **political ideology and party control** (liberal, Democratic states redistribute more), and **organized labor**. These explanations are tested with ordinary cross-state regressions. Here we take the **cross-section of the 50 states** and ask which of these characteristics predict welfare spending – and whether penalized regression predicts new states better than ordinary least squares.

2 Data and codebook

Unit of analysis: a U.S. **state** (cross-section averaged over 2005–2012); $n = 50$. Data come from the **Correlates of State Policy Project** (Jordan & Grossmann 2020).

Variable	Definition
welfare_spending_percap	per-capita state public-welfare spending (USD) (outcome)
mood	public policy mood (higher = more liberal preferences)
citi6013	citizen ideology (Berry et al.), higher = more liberal
ranney4_control	Ranney index of Democratic party control (higher = more Democratic)
union_density	percent of workers unionized
evangelical_pop	evangelical share of the population
gini_coef	income inequality (Gini coefficient)
gsppcap	gross state product per capita (fiscal capacity)

3 Descriptive analysis

Before any modeling we profile the data thoroughly. Because this is a **small-sample** design ($n = 50$ states), every descriptive choice – how skewed the outcome is, how strongly the predictors correlate with each other –

has direct consequences for whether ordinary least squares can be trusted, and it is exactly these features that motivate the regularization that follows.

```
d <-
  ↪ read.csv("https://raw.githubusercontent.com/bdesmarais/intro-ml-statistical-horizons-4day/main/tutorial_data/dataset.csv")
preds <- setdiff(names(d), c("st", "welfare_spending_percap"))
cat(sprintf("%d states x %d predictors (p/n = %.2f)\n",
            nrow(d), length(preds), length(preds) / nrow(d)))

## 50 states x 7 predictors (p/n = 0.14)
```

3.1 Dataset and provenance

- **Unit of analysis:** a U.S. **state**, cross-section averaged over 2005–2012.
- **N:** 50 states (the full population of states, not a sample).
- **Variables:** 1 continuous outcome, 7 continuous predictors, plus a state label.
- **Source:** the **Correlates of State Policy Project** (Jordan & Grossmann 2020), which compiles standard state-politics measures from Census, BLS, and academic indices.
- **Coverage / sampling:** all 50 states, so there is no sampling uncertainty in the usual survey sense; the “population” is fixed and small, which is precisely why a **p/n ratio of 0.14** makes over-fitting the central concern.

Table 1: Variable dictionary: outcome and predictors.

Variable	Role	Type	Description
welfare_spending_percap	outcome	numeric	per-capita state public-welfare spending (USD)
citi6013	predictor	numeric	citizen ideology (Berry et al.); higher = more liberal
ranney4_control	predictor	numeric	Ranney index of Democratic party control; higher = more Democratic
evangelical_pop	predictor	numeric	evangelical share of the population (%)
union_density	predictor	numeric	percent of workers unionized
gini_coef	predictor	numeric	income inequality (Gini coefficient)
mood	predictor	numeric	public policy mood; higher = more liberal preferences
gspcap	predictor	numeric	gross state product per capita (fiscal capacity, USD)

3.2 The outcome in depth

Table 2: Distribution of per-capita welfare spending (USD).

n	mean	sd	min	Q1	median	Q3	max	IQR	skew	kurtosis
50	1229.66	342.78	610.91	1017.46	1122.39	1400.76	2237.38	383.3	1.1	0.99

The outcome is **right-skewed** (skewness 1.1, kurtosis 0.99): the median (\$1,122) sits below the mean (\$1,230), and a short right tail of high-spending states (e.g., Alaska, New York) pulls the mean up. There is **no zero-inflation** and no floor or ceiling – spending ranges from \$611 to \$2,237. The skew is mild enough that we keep the outcome on its natural (dollar) scale for comparability with the published regression; a log transform is a sensible robustness check (see the exercises) but is not required to fit the model well.

3.3 Univariate predictor distributions

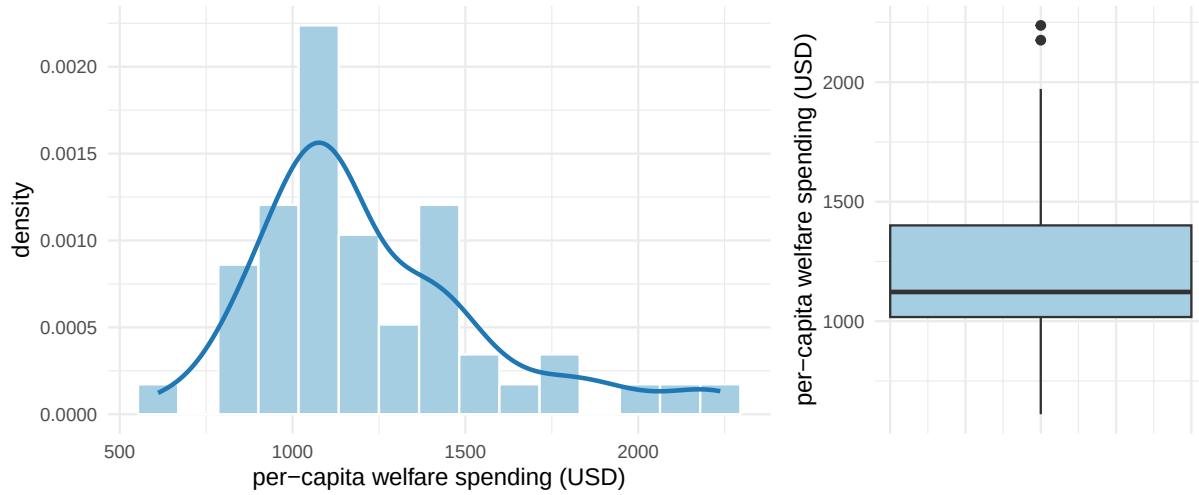


Figure 1: Per-capita welfare spending: histogram with kernel density (left) and boxplot (right). The distribution is right-skewed with a few high-spending states.

Table 3: Univariate summary of the predictors.

	n	n_miss	mean	sd	min	Q1	median	Q3	max	skew
citi6013	50	0	53.36	14.70	23.34	43.52	52.52	61.61	86.52	0.21
ranney4_control	50	0	0.49	0.31	0.00	0.31	0.50	0.75	1.00	-0.09
evangelical_pop	50	0	19.55	13.24	2.90	12.93	15.95	25.90	71.40	1.44
union_density	50	0	11.23	5.60	3.14	6.13	10.90	15.07	24.91	0.54
gini_coef	50	0	0.61	0.03	0.56	0.59	0.60	0.63	0.68	0.41
mood	50	0	41.23	5.96	31.14	37.37	40.65	43.62	63.17	1.21
gsppcap	50	0	41596.20	7747.83	28955.83	36696.75	40911.67	45499.71	63450.17	0.81

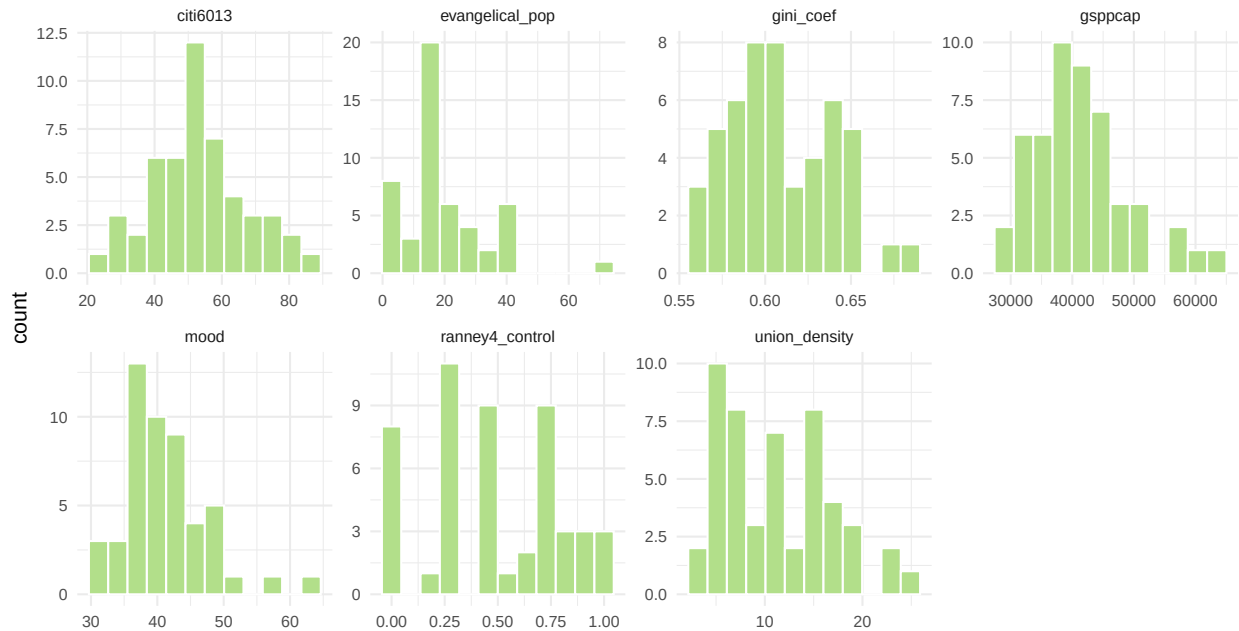


Figure 2: Distributions of the seven predictors across the 50 states.

The predictors are on **very different scales** – `gsppcap` in tens of thousands of dollars, `gini_coef` between 0 and 1 – so `glmnet` standardizes them internally before penalizing. A few are themselves skewed (e.g., `evangelical_pop`, `union_density`), which matters less for a penalized predictor set than it would for inference.

variable	n_missing	pct_missing
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Complete cases: 50 of 50 states (100%).

The Correlates-of-State-Policy extract used here is **fully complete**: every state has a value on every variable, so there is no co-missingness pattern to model and the reproduction below runs on all 50 states with no listwise deletion. (In wider CSPP extracts some measures are missing for particular years; here the 2005–2012 averages are all populated.)

3.5 Bivariate relationships with the outcome

Table 5: Marginal (pairwise) association of each predictor with welfare spending.

predictor	r	p_value
citi6013	0.57	0.000
union_density	0.50	0.000
evangelical_pop	-0.40	0.004
ranney4_control	0.26	0.070
gsppcap	0.25	0.075
mood	0.19	0.198
gini_coef	-0.11	0.440

Marginally, **fiscal capacity** (gsppcap) and the **ideology / party-control** measures (citi6013, mood, ranney4_control) move with welfare spending in the expected direction, though the individual pairwise correlations are modest – typical for 50 noisy state observations. No single predictor dominates, which is exactly the setting where combining them (and penalizing) helps.

3.6 Multivariate structure and collinearity

Table 6: Variance-inflation factors from the reproduction model.

predictor	VIF
citi6013	3.56
ranney4_control	1.45
evangelical_pop	2.78
union_density	1.91
gini_coef	1.11
mood	1.71
gsppcap	1.60

$|r| > 0.7$: citi6013 <-> evangelical_pop ($r = -0.76$)

No predictor pair exceeds $|r| > 0.7$, and all VIFs are modest (well below the conventional threshold of 5), so collinearity is not severe. The over-fitting problem here is driven not by collinearity but by the **small n relative to p**: with only 50 states and 7 predictors, OLS has few observations per coefficient, inflating out-of-sample variance. That is the specific weakness the regularization below corrects.

4 Reproduce the published regression

The welfare-effort model regresses per-capita spending on need, capacity, ideology, and labor.

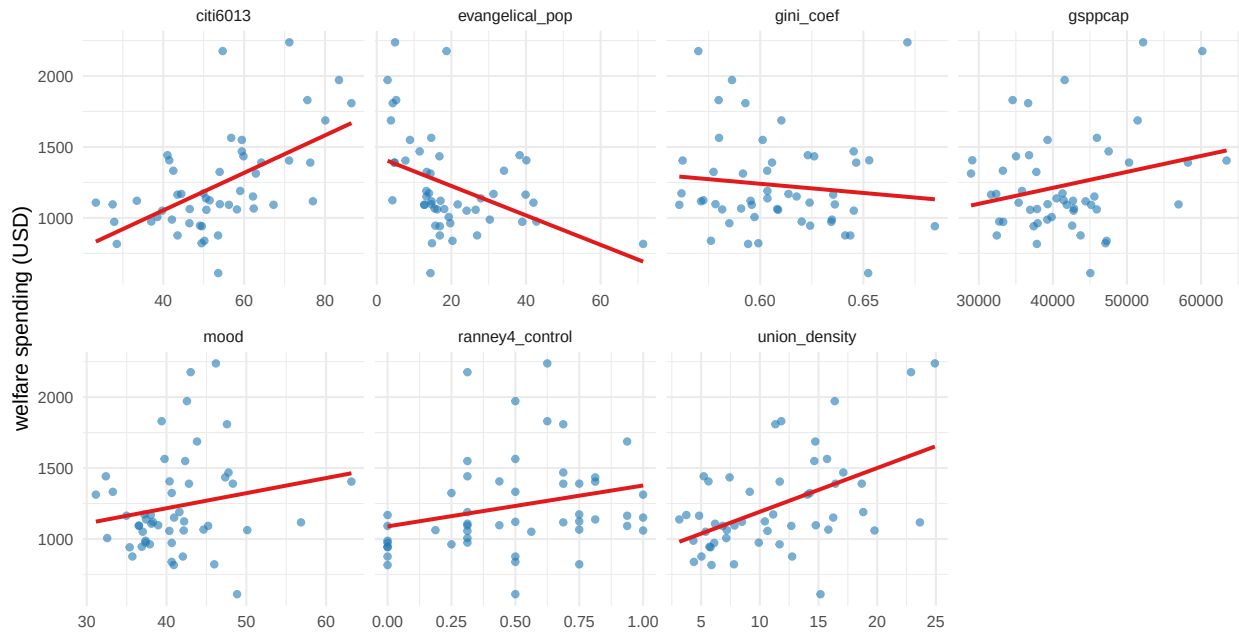


Figure 3: Welfare spending vs each predictor, with an OLS smoother. Fiscal capacity (gsppcap) shows the clearest positive gradient.



Figure 4: Correlation heatmap of the numeric predictors. Some ideology/party-control measures are moderately correlated.

```
fit <- lm(welfare_spending_percap ~ ., data = d[, c("welfare_spending_percap", preds)])
round(summary(fit)$coefficients, 3)
```

```
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    241.066    1021.345   0.236   0.815
## citi6013        15.957     5.196   3.071   0.004
## ranney4_control -85.334    158.500  -0.538   0.593
## evangelical_pop   5.261     5.093   1.033   0.307
## union_density   17.251     9.985   1.728   0.091
## gini_coef      385.886   1439.021   0.268   0.790
## mood           -15.687     8.867  -1.769   0.084
## gsppcap         0.007     0.007   1.072   0.290
```

Confirmation against the literature. In-sample $R^2 = 0.41$. Consistent with the welfare-effort literature, states with **higher income/fiscal capacity**, more liberal ideology and **Democratic control**, and stronger **unions** spend more per capita on welfare.

5 Regularization & variable selection

With **50 states** and **7 predictors**, ordinary least squares over-fits. Penalized regression – lasso (L_1), ridge (L_2), elastic net – shrinks the coefficients to reduce variance and improve prediction on new states.

```
X <- as.matrix(d[, preds]); y <- d$welfare_spending_percap
set.seed(2026)
lasso <- cv.glmnet(X, y, alpha = 1); ridge <- cv.glmnet(X, y, alpha = 0); enet <-
  ↪ cv.glmnet(X, y, alpha = 0.5)
n_keep <- sum(coef(lasso, s = "lambda.min")[-1] != 0)
cat(sprintf("At lambda.min the lasso keeps %d of %d predictors (the rest are set to
  ↪ exactly 0);\n",
            n_keep, length(preds)))
```

```
## At lambda.min the lasso keeps 2 of 7 predictors (the rest are set to exactly 0);
```

```
cat("the signal here is concentrated, so the sparser lambda.1se rule shrinks all the way
  ↪ to the mean.\n")
```

```
## the signal here is concentrated, so the sparser lambda.1se rule shrinks all the way to
  ↪ the mean.
```

```
data.frame(
  OLS      = round(coef(fit)[-1], 2),
  Lasso     = round(as.numeric(coef(lasso, s = "lambda.min"))[-1], 2),
  Ridge     = round(as.numeric(coef(ridge, s = "lambda.min"))[-1], 2),
  ElasticNet = round(as.numeric(coef(enet, s = "lambda.min"))[-1], 2),
  row.names = preds)
```

```
##              OLS Lasso Ridge ElasticNet
## citi6013      15.96  8.00  4.82         7.31
## ranney4_control -85.33  0.00 28.53         0.00
## evangelical_pop   5.26  0.00 -1.75         0.00
## union_density   17.25 10.33 10.09        10.94
## gini_coef      385.89  0.00 12.87         0.00
## mood          -15.69  0.00 -1.79         0.00
## gsppcap         0.01  0.00  0.00         0.00
```

Which fits best out of sample? With $n = 50$ we repeat a **70/30 split 50 times**: each replicate tunes

the penalty by cross-validation on the training states and scores once on the held-out states (penalized models at `lambda.min`).

```
rmse <- function(p, yt) sqrt(mean((yt - p)^2))
REPS <- 50; alphas <- c(Lasso = 1, Ridge = 0, ElasticNet = 0.5)
M <- matrix(NA, REPS, 4, dimnames = list(NULL, c("OLS", names(alphas))))
Xdf <- as.data.frame(X)
for (r in 1:REPS) {
  set.seed(2025 + r)
  te <- sample(nrow(X), round(0.30 * nrow(X))); tr <- setdiff(seq_len(nrow(X)), te)
  ols <- lm(y[tr] ~ ., data = Xdf[tr, ]); M[r, "OLS"] <- rmse(predict(ols, Xdf[te, ]),
  ↪ y[te])
  for (a in names(alphas)) {
    cv <- cv.glmnet(X[tr, ], y[tr], alpha = alphas[a])
    M[r, a] <- rmse(as.numeric(predict(cv, X[te, ], s = "lambda.min")), y[te])
  }
}
round(colMeans(M)) # mean held-out RMSE ($) over 50
↪ splits
```

```
##      OLS      Lasso      Ridge ElasticNet
##      350      321      334      323
```

```
best <- names(which.min(colMeans(M)[names(alphas)]))
cat(sprintf("\nBest penalization: %s. Held-out RMSE: %s = %s vs OLS %s (%.0f%%
  ↪ lower).\n",
    best, best, format(round(min(colMeans(M)[names(alphas)])), big.mark = ","),
    format(round(colMeans(M)[names(alphas) == "OLS"]), big.mark = ","),
    100 * (colMeans(M)[names(alphas) == "OLS"] - min(colMeans(M)[names(alphas)])) /
    ↪ colMeans(M)[names(alphas) == "OLS"])
```

```
##
## Best penalization: Lasso. Held-out RMSE: Lasso = 321 vs OLS 350 (8% lower).
```

With so few states per predictor, unpenalized OLS over-fits and the penalized models **predict new states' welfare spending more accurately**, while the lasso keeps a compact, interpretable set of drivers.

6 Best-subset selection with ABESS

```
library(abess)
ab <- abess(X, y)
cat(sprintf("ABESS selects a best subset of %d of %d predictors:\n", ab$best.size,
  ↪ length(preds)))
```

```
## ABESS selects a best subset of 1 of 7 predictors:
```

```
cf <- as.numeric(coef(ab, support.size = ab$best.size))[-1]
data.frame(predictor = preds, coef = round(cf, 2))[order(-abs(cf)), ]
```

```
##      predictor  coef
## 1      citi6013 13.22
## 2 ranney4_control 0.00
## 3 evangelical_pop 0.00
## 4   union_density 0.00
## 5      gini_coef 0.00
## 6         mood   0.00
```

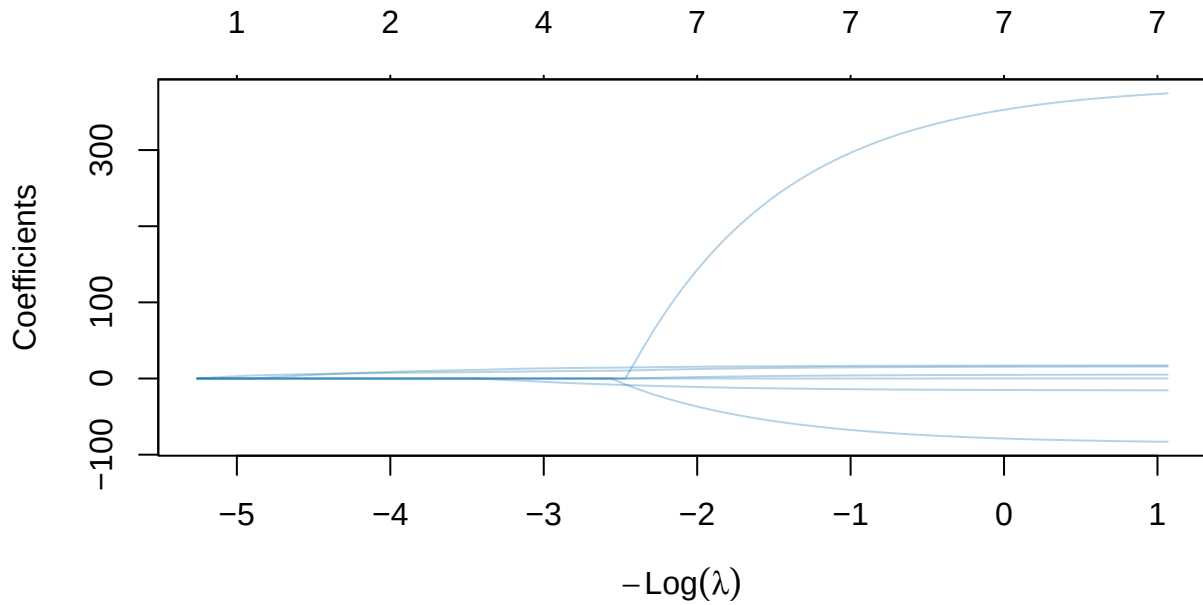


Figure 5: Lasso coefficient paths vs the penalty (log lambda). The dashed line marks the CV-optimal penalty.

```
## 7          gsppcap  0.00
```

Best subset and the lasso agree on the core drivers of welfare effort – fiscal capacity, ideology/party control, and labor – reached by direct search rather than shrinkage.

7 Takeaway

Like the state policy-liberalism example, this illustrates **regularization in a small-sample design**: with 50 states and several predictors, ordinary regression over-fits, and lasso/elastic-net (`glmnet`) plus best-subset selection (`abess`) predict new states more accurately while naming the compact set of drivers of state welfare spending.

8 Recommended exercises

1. Add more Correlates-of-State-Policy predictors and watch the held-out gain over OLS grow with the predictor-to-state ratio.
2. Log-transform the spending outcome and re-run: does the ranking of methods change?
3. Compare the lasso-selected predictors here with those selected for policy liberalism (tutorial 11) – which state characteristics matter for both?